

TGR-LoRA: Task-Guided Gradient-Reset Low-Rank Adaptation for Continual Learning in Large Language Models

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Motivation

Fine-tuning LLMs on a sequence of tasks with LoRA leads to **catastrophic forgetting**: gradient updates for the new task *overwrite* the parameter directions learned by earlier tasks. Current PEFT methods such as LoRA often treat tasks in isolation or overwrite parameters learned from previous tasks, failing to explore potential cross-task knowledge-reuse and facing the problem of catastrophic forgetting as well as significant storage overhead.

Project Objectives

TGR-LoRA is designed to satisfy the following three core goals:

- Zero Forgetting**: Ensures zero forgetting of previously learned tasks through strict isolation via orthogonal subspaces.
- Forward Transfer**: Explicitly leverages previously acquired knowledge via a Mixing Matrix to accelerate adaptation to new tasks.
- Backward Transfer**: Potentially optimizes and enhances the performance of earlier tasks through the evolution of sharing mechanisms.

TRACE Benchmark

Eight sequential tasks including diverse domains, evaluated on **Qwen2.5-3B-Instruct**:

Task	Metric
C-STANCE (Chinese stance)	Accuracy
FOMC (financial)	Accuracy
ScienceQA (science reasoning)	Accuracy
NumGLUE-cm (math calculation)	Exact Match
NumGLUE-ds (data science math)	Exact Match
MeetingBank (meeting summarisation)	ROUGE-L
20Minuten (text simplification)	SARI
Py150 (code completion)	Edit Sim.

Tasks arrive one at a time; the model must retain all prior performance while adapting to each new task/domain.

TGR-LoRA vs O-LoRA

	O-LoRA	TGR-LoRA
Orthogonal grad. proj.	✓	✓
Frozen basis pool	×	✓
Trainable reuse (M_t)	×	✓
Pool grows with tasks	×	✓

O-LoRA only prevents forgetting. TGR-LoRA additionally *leverages* historical subspaces through the low-rank Mixing Matrix $M_t = U_t V_t^\top$.

TGR-LoRA Algorithm

Algorithm 1: TGR-LoRA Continual Learning

Input: Tasks $\{T_1, \dots, T_N\}$; LoRA rank r ; scaling α ; mt-rank k

Output: Model + frozen Basis Pool

```
for each task  $t = 1$  to  $N$  do
  // Step 1 -- Initialize Active Zone
   $A_t \sim$  Kaiming-uniform
  if  $t = 1$  then
     $B_t \leftarrow \mathbf{0}$  // standard LoRA cold start
  else
    Sample  $\tilde{B} \sim \mathcal{N}(\mathbf{0}, I)$ 
     $B_t \leftarrow \tilde{B} - Q_{\text{pool}}(Q_{\text{pool}}^\top \tilde{B})$ 
     $B_t \leftarrow \text{QR}(B_t)$  // init orthogonal to all pool bases
  end
  // Step 2 -- Initialize Mixing Matrix  $M_t$ 
   $r_{\text{tot}} \leftarrow (t-1) \cdot r$ 
   $U_t, V_t \in \mathbb{R}^{r_{\text{tot}} \times k}$ , zero-pad from  $M_{t-1}$  (warm start)
  // Step 3 -- Training Loop
  while not converged do
    Forward:  $\hat{y} = W_0 x + \Delta W_t(x)$ 
    Backward, then project gradient:
     $g \leftarrow \nabla_{B_t} \mathcal{L}$ 
    for  $j = 1$  to  $t-1$  do
       $g \leftarrow g - Q_j(Q_j^\top g)$  // Gram-Schmidt onto  $\text{span}(Q_j)^\perp$ 
    end
    Apply projected gradient  $g$  to  $B_t$ 
    Update  $A_t, U_t, V_t$  (no projection)
  end
  // Step 4 -- Commit to Basis Pool
   $[Q_t, R_t] \leftarrow \text{QR}(B_t)$  //  $Q_t^\top Q_t = I$ ; lossless factorisation
   $C_t \leftarrow R_t A_t$  //  $Q_t C_t x = B_t A_t x$  preserved exactly
  Freeze  $(Q_t, C_t) \rightarrow$  Basis Pool
  Freeze and archive  $M_t$ 
  Reset:  $B_t \leftarrow \mathbf{0}$ ,
   $A_t \sim$  Kaiming
end
```

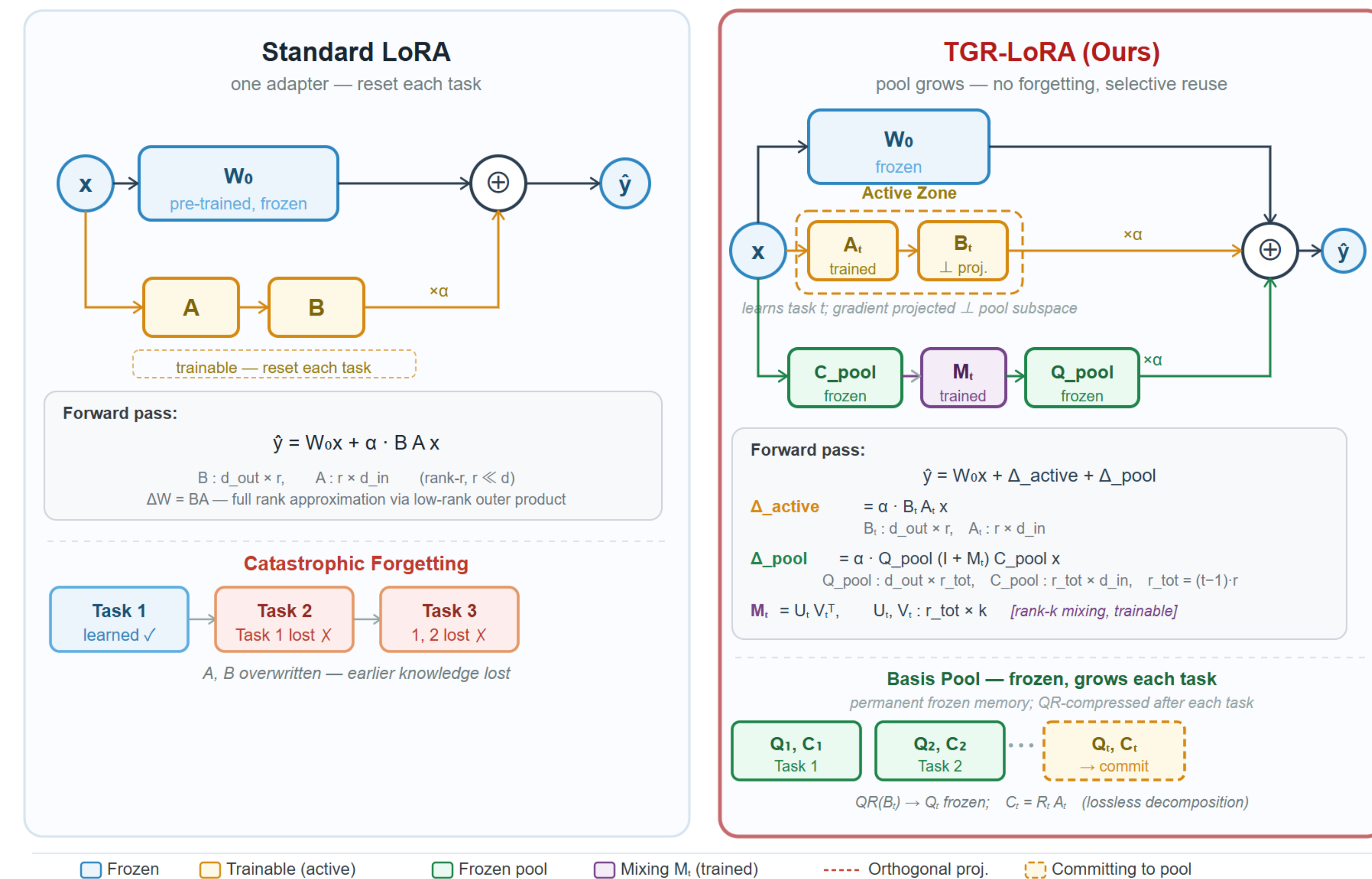


Figure 1. Architecture overview. The Basis Pool (grey, frozen) grows by one block per task.

Experimental Results

Setup: Qwen2.5-3B-Instruct, rank $r=32$, $\alpha=64$, mt-rank $k=4$, 3 epochs/task.

After Task	Average Acc $\uparrow$	BWT $\uparrow$
C-STANCE (Task 1)	0.370	0.000
FOMC (Task 2)	0.815	+0.320

Observation: After training on Task 2 (FOMC), the model not only masters the new domain but significantly enhances its performance on Task 1, indicating powerful knowledge synergy through the Mixing Matrix.

Key Finding: Positive Backward Transfer

TGR-LoRA achieves a rare **Positive Backward Transfer (BWT = +0.32)**. While standard LoRA suffers from forgetting and O-LoRA merely maintains performance, TGR-LoRA retroactively improves the earlier task:

	C-STANCE Acc	FOMC Acc
After Task 1	0.37	—
After Task 2	0.69	0.94
Improvement	+86%	—

This improvement occurs because the FOMC training gradients, although orthogonal to the C-STANCE basis, optimize the **Mixing Matrix M_t** to better leverage shared sentiment features between the two domains.

Knowledge Reuse Analysis

The Mixing Matrix M_t quantitatively encodes cross-task reuse:

$$\text{Reuse}(t \rightarrow i) = \|M_t[\text{block } i]\|_F$$

Our analysis shows that FOMC (Financial Sentiment) actively reuses the C-STANCE (Social Media Stance) subspace. This aligns with the semantic similarity: both tasks require identifying underlying attitudes and sentiment polarities within text, allowing the model to "warm-start" from historical features.

Conclusion & Future Work

TGR-LoRA contributes three integrated mechanisms:

- Orthogonal isolation** via gradient projection into $\text{span}(Q_{\text{pool}})^\perp$, eliminating catastrophic forgetting.
- Lossless retention** via a QR-compressed Basis Pool that grows as the model encounters new domains.
- Cross-task transfer** via the low-rank Mixing Matrix M_t , turning task interference into task synergy.

Future directions: Full 8-task evaluation; adaptive rank selection for M_t ; scaling to 7B+ parameters.

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